



All solution-processed SnO_2 /1D- CsAg_2I_3 heterojunction for high-sensitivity self-powered visible-blind UV photodetector

Zhaosheng Hu^{1,2}, Boyao Zhang¹, Feijuan Zhang¹, Xing Guo¹, Zhenhua Lin^{1,2}, Jincheng Zhang¹, Yue Hao¹ and Jingjing Chang^{1,2*}

ABSTRACT Low-dimensional lead-free metal halides have shown considerable potential for visible-blind ultraviolet (UV) photodetectors (PDs) owing to their extraordinary optoelectronic properties. Herein, a high-quality perovskite-like CsAg_2I_3 thin film was successfully fabricated *via* the pyridine additive-assisted one-step solution method, and it exhibited an intrinsic p-type behavior and possessed a low exciton binding energy of 202 meV, which was even comparable to low-dimensional lead halide perovskite counterpart and thus favored the separation of photogenerated carriers. Additionally, a type-II n-p heterojunction of SnO_2 / CsAg_2I_3 for a self-powered UV PD was configured and investigated experimentally and theoretically. A 4.13-eV-wide-bandgap SnO_2 with an extremely low Urbach energy of 0.214 eV was obtained to suppress the persistent photoconductivity effect. The resulting device exhibited pronounced sensitivity with an ultrafast response time of 47/74 μs , which is outstanding among visible-blind UV PDs and two orders of magnitude lower than previously reported CsAg_2I_3 -based UV PDs. The responsivity and detectivity were as high as 0.032 A W^{-1} and 1.2×10^{11} Jones, respectively. Furthermore, the highly transparent (> 90%) heterojunction for visible light illustrated the superior visible-blind characteristics. The remarkable device performance, solution-processability, and long-term stability reveal a promising strategy for the design of commercial UV PDs that are visible-blind, self-powered, stable, and nontoxic.

Keywords: CsAg_2I_3 , UV photodetector, visible-blind, transparent, perovskite-like

INTRODUCTION

Ultraviolet (UV) photodetectors (PDs) are crucial in various applications, such as imaging systems, environmental surveillance, communication and biological sensing [1–3]. In the past years, a series of UV PDs based on wide-bandgap metal oxides and nitrides such as SnO_2 [4,5], ZnO [6,7], TiO_2 [8,9], and GaN [10,11] have been investigated. However, they are frequently applied in the form of nanostructures such as single millimeter wires, nanowire arrays, nanobelts, and nanofibers, and the involved fabrication process is complicated for industrial applications. Additionally, the growth of these structures for high-performance UV PDs requires vacuum equipment such as molecular beam epitaxy, magnetron sputtering, and pulsed laser

deposition [12–16], which makes the UV PDs expensive. Metal halide materials, particularly lead halide perovskites, have attracted intensive attention because of their solution-processability and excellent optoelectrical properties. The incorporation of chloride opens up the possibility of obtaining wide-bandgap materials. MAPbCl_3 and CsPbCl_3 have been used for UV PDs [17,18]. However, the toxicity of lead limits their further industrial applications. Moreover, the chloride anion makes their preparation challenging using solution-processed methods. To achieve high-performance visible-blind UV PDs, a stable ecofriendly wide-bandgap material is highly desired. Bismuth halides such as double perovskites are widely studied materials, but they are not suitable for UV PDs because of their indirect bandgaps, which inhibit light absorption and conversion. As low-dimensional materials usually yield a wider bandgap than their counterpart 3D materials, inorganic silver or copper halides such as zero-dimensional (0D) $\text{Cs}_3\text{Cu}_2\text{I}_5$ [19] and 1D CsCu_2I_3 [20] are studied and applied in UV PDs. All of them are stable and neatly blind to > 405 nm visible light. Additionally, the investigation of 1D CsAg_2I_3 [21] reveals a direct wide bandgap with ~365 nm absorption edge. Recently, a photoresponsivity and response time of 23.59 mA W^{-1} and 6/5.4 ms, respectively, were reported for CsAg_2I_3 [22]. However, UV PDs based on these low-dimensional lead-free metal halides still suffer from slow photoresponse, which needs to be further addressed.

Herein, we synthesized a high-quality perovskite-like inorganic CsAg_2I_3 thin film through a solution-processed method. The film exhibited a low exciton binding energy comparable to its low-dimensional lead halide perovskite counterpart. Additionally, a highly sensitive self-powered visible-blind UV PD based on the typical type-II heterojunction of SnO_2 /1D- CsAg_2I_3 was achieved. Wide-bandgap SnO_2 with extremely low Urbach energy was obtained to suppress the persistent photoconductivity (PPC) effect. The planar heterojunction not only enabled ultrafast photoresponse, high responsivity and detectivity, and remarkable stability, but also exhibited high transparency under radiation with > 360 nm wavelength, revealing superior visible-blind characteristics.

EXPERIMENTAL SECTION

Fabrication of CsAg_2I_3 films and devices

CuI , AgI , dimethyl formamide (DMF), and dimethylsulfoxide

¹ State Key Discipline Laboratory of Wide Band Gap Semiconductor Technology, School of Microelectronics, Xidian University, Xi'an 710071, China

² Advanced Interdisciplinary Research Center for Flexible Electronics, Xidian University, Xi'an 710071, China

* Corresponding author (email: jingchang@xidian.edu.cn)

(DMSO) were purchased and used without any further purification. First, a mixture of CsI and AgI with a molar ratio of 1:2 was dissolved in DMSO to prepare the precursor solution. Then, 100 μL pyridine was added to the solution and stirred for a whole night and kept at 70°C in a glovebox. The indium tin oxide (ITO) substrates were sequentially washed with deionized water and isopropanol for 30 min and dried by using a hot plate before the deposition of thin films. A compact SnO_2 film was deposited on the substrate *via* spin-coating (4000 r min^{-1} , 30 s) and annealing (150°C for 30 min) in air. It is worth noting that pre- and post-UV ozone treatment for ~ 5 min was utilized to suppress the oxygen vacancy. Then, the CsAg_2I_3 precursor solution of $\sim 80 \mu\text{L}$ was dropped on the substrate and spin-coated at 500 r min^{-1} for 10 s and 2000 r min^{-1} for 30 s. Afterward, the gradient thermal annealing method (first at 50°C for 5 min and then at 100°C for 1 h) was immediately performed. To complete the device fabrication, a 100-nm silver electrode was deposited *via* thermal evaporation.

Characterization

The crystal structures were characterized using X-ray diffraction (XRD) with Cu K α radiation (Bruker D8 Advance). Scanning electron microscopy (SEM) images were obtained using an SEM FEI-Apreo. The absorbance spectrum was obtained using a UV-visible (UV-vis) spectrophotometer (Hitachi UH4150). The photoluminescence (PL) spectrum was recorded using a fluorescence spectrophotometer FLS1000 with a 340-nm picosecond pulsed laser. The energy level was evaluated using UV photoelectron spectroscopy (UPS; PHI 5000 VersaProbe III with a He I source [21.22 eV]). The photocurrent measurements were conducted under ambient conditions at room temperature using Keithley 2450. External quantum efficiency (EQE) measurements were conducted using the SCS10-X150 systems (Zolix Instrument Co., Ltd.). The dark current was recorded using a semiconductor setup (4200-SCS, Keithley Co.).

Theoretical calculations

The electric field distribution in the device was simulated *via* the finite-difference time-domain method using Meep software package developed at MIT (Massachusetts Institute of Technology). Furthermore, a 2D modeling was performed on the Gauss source. An exact full-field electromagnetic calculation based on Maxwell's equations in the frequency domain was performed with perfectly matched layer boundary conditions.

Elliot model, Urbach tail, and Mott-Gurney fitting: Considering the excitonic effect, the UV-vis absorption spectrum can be described using Elliott's formula [23,24],

$$\alpha(E) = \frac{A}{E} \sum_n \left[\frac{2E_b^{1.5}}{n^3} \delta(E - E_g^b) + \frac{\sqrt{\frac{E_b}{E - E_g}} \Theta(\sqrt{E - E_g})}{1 - \exp\left(-2\pi \sqrt{\frac{E_b}{E - E_g}}\right)} \right] \quad (1)$$

where A is a constant, $E_g^b = E_g - E_b/n^2$, $\Theta(x)$ is the step function, E is the photon energy, E_b is the exciton binding energy, and E_g is the band gap. Gaussian broadening functions were convoluted to fit the experimental data. The Urbach tail fitting follows the following relation [25]:

$$\alpha \propto \exp\left(\frac{E - E_{\text{on}}}{E_U}\right), \quad (2)$$

where E_{on} is the energy onset of the tail, and E_U is the Urbach energy quantifying the energetic disorder in the semiconductor. The charge carrier mobility can be estimated in the space-charge-limited current (SCLC) regime as follows [26,27]:

$$J = \frac{9\varepsilon_0\varepsilon_r\mu V^2}{8L^3}, \quad (3)$$

where ε_0 , ε_r , μ , V , and L are the permittivity of free space, the dielectric constant of the active layer, carrier mobility, the voltage drop across the device, and the active layer thickness, respectively.

RESULTS AND DISCUSSION

The CsAg_2I_3 thin film was prepared *via* a one-step solution method, and pyridine was introduced as an additive to improve crystallization. In our previous study [28], the additive was found to possess strong chemical bonding to metal cations and induce high-quality lead-free bismuth-based halide films. The as-prepared film (Fig. 1a) is smooth, compact, and transparent without any pinholes. The flat ribbon-like grains with large grain size and few boundaries were observed, which benefits carrier transport. CsAg_2I_3 was characterized *via* XRD. As shown in Fig. 1b, the sharp diffraction peaks at 10.34°, 21.08°, 31.63°, and 42.45° correspond to the (110), (130), (022) and (242) crystal planes, and the relatively weak peaks at 13.30° and 25.51° are (020) and (131), respectively, consistent with previous results [21]. The marked weak XRD peaks are attributed to the substrate, consistent with our previous report [28]. The CsAg_2I_3 features an orthorhombic unit cell with the space group $Pbnm$ (Fig. 1c). To confirm the electronic band structure of CsAg_2I_3 , UV-vis spectroscopy and UPS were performed. As shown in Fig. 1d, the CsAg_2I_3 film exhibits strong absorption for UV illumination with an obvious cut-off edge at ~ 350 nm. This is mainly due to the wide direct bandgap (3.54 eV) of the film, estimated according to Elliott fitting [23,24], as the conventional Tauc analysis neglects the exciton contribution near the band edge. The bandgap value is ~ 0.1 eV larger than a previously reported bulk single crystal [21], attributable to the influence of thickness on localized states. Additionally, the estimated exciton binding energy of CsAg_2I_3 is 202 meV, which is considerably larger than the value of typical 3D lead halide perovskites (~ 30 meV) but comparable to low-dimensional perovskites such as 2D materials $(\text{PEA})_2\text{PbI}_4$ (239 meV) and $(\text{BA})_2\text{PbI}_4$ (470 meV) [29,30]. The relatively low binding energy of the 1D CsAg_2I_3 is beneficial for the separation of photogenerated carriers.

The Fermi level (E_f) and valence band maximum (E_v) were estimated *via* UPS using a He I source (21.22 eV). As shown in Fig. 1e, the E_f and E_v of the CsAg_2I_3 films are -5.69 and -7.37 eV, respectively. The conduction band minimum (E_c) evaluated according to the difference between E_v and E_g is -3.83 eV. The $E_f - E_v$ is 1.68 eV, which is smaller than half of the bandgap. Hence, the as-grown CsAg_2I_3 is p-type, which is suitable for forming an all-wide-bandgap p-n heterojunction with n-type metal oxides. Additionally, the PL spectra (Fig. 1f, g) reveal that CsAg_2I_3 exhibits a PL peak at 602 nm, possessing a large Stokes shift due to the frequently observed self-trapped states in low-dimensional materials [21,31]. The PL decay curve shows an average PL lifetime of 3.2 ns, which is shorter than bulk or single-crystal CsAg_2I_3 and can be beneficial for the fast response of UV PDs from an application point of view [21]. The electron and hole mobilities determined according to the Mott-

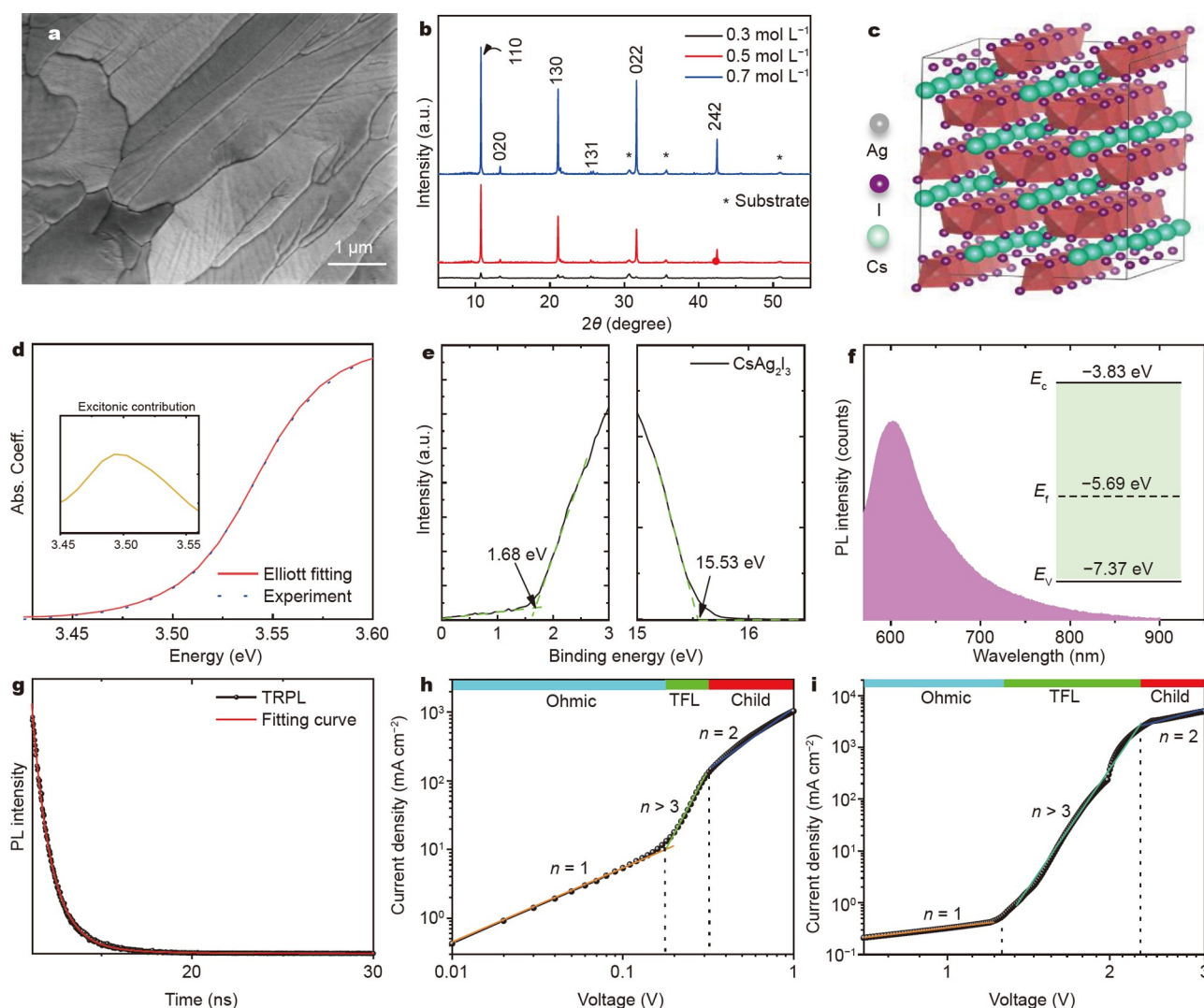


Figure 1 (a) SEM image and (b) XRD patterns (as a function of the precursor concentration) of the CsAg_2I_3 thin films. (c) Crystal cell of CsAg_2I_3 with an orthorhombic structure. (d) Elliott model fitting of the absorption coefficient spectrum. (e) UPS spectra obtained using a He I source (21.22 eV). (f) PL (the inset shows the electronic structure of CsAg_2I_3) and (g) time-resolved PL spectrum with the fitting curve of CsAg_2I_3 . (h, i) SCLC of the electron- and hole-only devices.

Gurney law [26] from the SCLC approach are 0.098 and $0.145 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, respectively (Fig. 1h, i). Similar to the exciton binding energy, the carrier mobility of the as-prepared CsAg_2I_3 is lower than 3D perovskites but analogous to low-dimensional materials such as $\text{Cs}_3\text{Cu}_2\text{I}_5$ [19], probably because the binding energy has a negative influence on carrier mobility.

As heterojunction engineering is an important strategy for promoting the extraction of photogenerated carriers and forming self-powered PDs [32,33], n-type wide-bandgap SnO_2 was combined with CsAg_2I_3 (p-type) to form an n-p heterojunction. Among n-type metal oxides, SnO_2 possesses high electron mobility, stability, and visible light transparency [34]. Moreover, polycrystalline films can be fabricated at low temperatures and are frequently used in flexible photovoltaics [35]. As shown in Fig. S1, the solution-processed thin-film SnO_2 is uniform, compact, and smooth. To suppress the widely existing PPC effect in wide-bandgap semiconductors, particularly for metal oxides and nitrides, which may affect the response time of the PDs [36], annealing in air after spin-coating of SnO_2 and suffi-

cient UV pre/post-treatment were adopted to suppress the oxygen vacancy during fabrication. Compared with the reported SnO_2 [37,38] (Fig. S2), a SnO_2 film with 4.13-eV-wide-bandgap and an extremely low Urbach energy of 0.214 eV, estimated from the Tauc plot and Urbach tail, was obtained. Such a low Urbach energy is beneficial for photoresponse. Fig. 2a, b show the constructed vertical device configuration and the energy band diagram. SnO_2 and CsAg_2I_3 form a typical type-II contact at the interface, accelerating the separation and transportation of photogenerated carriers, and the built-in electric field inside the depletion region allows the device to work under zero voltage bias as an effective self-powered UV PD.

To optimize the constructed device structure, the film thicknesses of CsAg_2I_3 and SnO_2 were varied by tuning the corresponding precursor concentration (Fig. S3 inset). The optoelectronic characteristics of the as-prepared devices were characterized. Fig. S3a shows the measured EQE, which is defined as the ratio of the output number of electrons/holes to the number of incident photons. The devices are pronouncedly

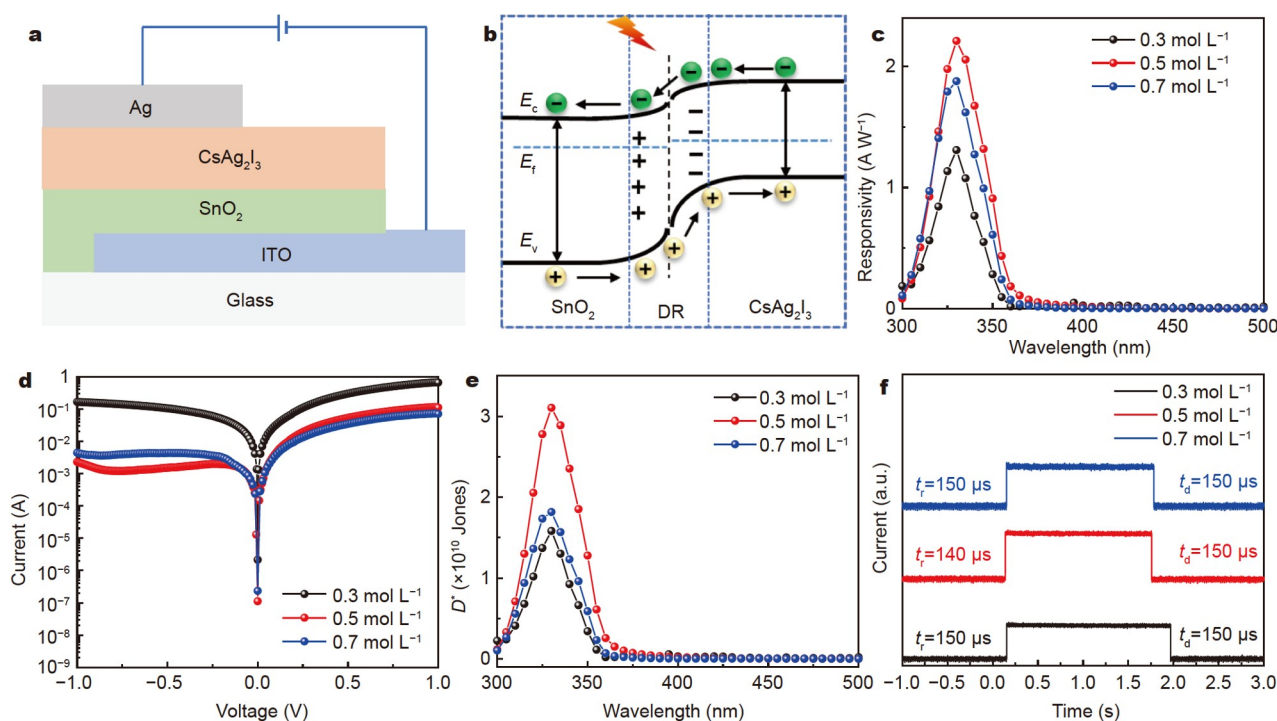


Figure 2 Photoelectric performances of $\text{SnO}_2/\text{1D-CsAg}_2\text{I}_3$ UV PDs. (a) Device structure of the UV PD. (b) Energy band diagram of the $\text{SnO}_2/\text{CsAg}_2\text{I}_3$ heterojunction. (c) Responsivity, (d) dark current-voltage, (e) detectivity, and (f) response time of $\text{SnO}_2/\text{1D-CsAg}_2\text{I}_3$ UV PDs as a function of the CsAg_2I_3 precursor concentration.

sensitive to UV illumination with < 350 nm wavelength, mainly due to the bandgap of CsAg_2I_3 . The climax EQE value was obtained at 330 nm. For CsAg_2I_3 , the precursor concentration of 0.5 mol L^{-1} was confirmed to be the optimum value, and the EQE reached 8.31% under 330 nm illumination. In addition to EQE, responsivity (R) and specific photodetectivity (D^*) are two key parameters for quantitatively evaluating PD performance and reflecting the response of the PD to the incident light and the smallest detectable signals of the PDs from the noise environment, respectively [39,40]. They can be estimated using the following equations [21]:

$$R = \frac{I_{\text{light}} - I_{\text{dark}}}{P_{\text{light}} S}, \quad D^* = \frac{R}{(2qJ_{\text{dark}})^{1/2}}, \quad (4)$$

where I_{light} , I_{dark} , P_{light} , S , q , and J_{dark} are the photocurrent, dark current, incident light intensity, the effective size of PD, elementary charge, and dark current density, respectively.

Fig. 2c, d show that R and D^* have similar trends, and the values for $\text{SnO}_2/\text{CsAg}_2\text{I}_3$ PDs under 330 nm light reach $\sim 0.021 \text{ A W}^{-1}$ and $\sim 3.11 \times 10^{10}$ Jones, respectively. The high performance under a 0.5 mol L^{-1} precursor concentration is attributable to the relatively low dark current ($\sim 1.1 \times 10^{-7} \text{ A}$) of the PD under zero voltage bias (Fig. 2e). The UV/visible rejection ratio, defined as the ratio of the responsivities at 330 and 400 nm (R_{330}/R_{400}), is $\sim 1.9 \times 10^2$, confirming the main response of the PD to visible-blind signals. Moreover, the lowest dark current always occurs around zero bias, contributing to the high photodetectivity of the self-powered UV PDs.

Response time is a crucial parameter in UV detection. UV PDs are widely used as industrial optoelectronic sensors. Fast response is desperately in demand for signal acquisition to distinguish varied UV lights with high frequency. In this work, the

devices were operated under zero voltage bias to determine the response time. As shown in Fig. 2f, the rising time t_r (time taken for the photocurrent to increase from 10% to 90% of its steady state value) and decaying time t_d (time required for the photocurrent to decay from 90% to 10% of its steady state value) are approximately 140 and 150 μs , respectively, while the rising and decaying time constants (τ_r and τ_d) calculated *via* exponential curve fitting are 64 and 69 μs , respectively. The strong dependence on CsAg_2I_3 film thickness is mainly due to the bandgap arrangement in the device. Although SnO_2 was fronted by the incident illumination, almost all incident radiation with an operational wavelength (300–356 nm) was absorbed by CsAg_2I_3 owing to the short cut-off wavelength (~ 300 nm, 4.13 eV) of SnO_2 . On the other hand, the relatively low mobility of CsAg_2I_3 usually leads to a short diffusion length, which in turn restricts its optimum film thickness in the device, considering the recombination of photogenerated carriers.

In addition to CsAg_2I_3 , the photoresponse of the UV PDs was found to strongly depend on the film thickness of SnO_2 . Fig. S4 and Fig. 3a–c show the corresponding R , D^* , response time, and dark current. EQE is provided in Fig. S3b. All merits for various SnO_2 thicknesses reveal a similar dependence on the incident light wavelength, with the climax at 330 nm. The precursor concentration of 4 wt% yields outstanding device performance. The EQE and R under 330 nm illumination are 12.3% and 0.032 A W^{-1} , respectively. Moreover, the dark current decreases to $\sim 10^{-8} \text{ A}$. The τ_r/τ_d and t_r/t_d times are 22/34 and 47/74 μs , respectively. The rise and decay edges are provided in Fig. S5. Besides, the detectivity reaches 1.2×10^{11} Jones. Compared with the recent reports on visible-blind UV PDs (Table 1 [19,21,22,41–46]), the response time in this work is outstanding; it is two orders of magnitude lower than that of the latest report

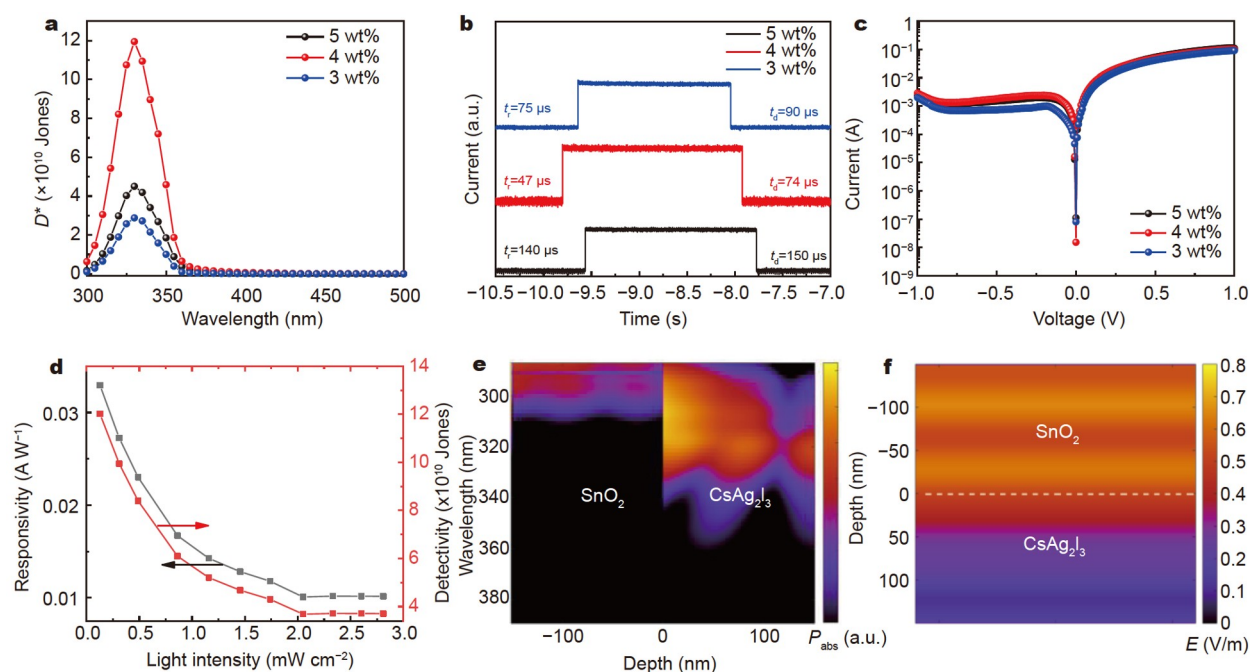


Figure 3 (a) Detectivity, (b) response time, and (c) dark current-voltage of the UV detector as a function of the SnO_2 precursor concentration. (d) Dependence of light intensities on responsivity and detectivity. (e) Power absorption per unit volume (P_{abs}) and (f) electric field distribution of the $\text{SnO}_2/\text{CsAg}_2\text{I}_3$ heterojunction.

Table 1 Comparison of the performances of visible-blind UV (< 400 nm) PDs based on different photoactive materials

UV PD structures	Wavelength (nm)	R (A W^{-1})	D^* (Jones)	Rise/decay time	Ref.
2D ZnO nanosheets	254	2.0×10^4	6.83×10^{14}	3.97/5.32 s	[41]
AlGaIn/GaN 2DEG-IPDs	365	1670	—	0.05/0.36 s	[42]
In/GaN/CsCu ₂ I ₃ /gold	275	0.047	2.58×10^{12}	2.6/2.7 ms	[43]
Gr/PtSe ₂ /β-Ga ₂ O ₃	245	0.076	1.93×10^{13}	12/212 μs	[44]
CsAg ₂ I ₃	265	3.55×10^{-4}	2.95×10^9	23/43 ms	[21]
CsAg ₂ I ₃ single crystal	310	0.0236	2.81×10^{10}	6/5.4 ms	[22]
Cs ₃ Cu ₂ I ₅	265	0.0649	6.9×10^{11}	26.2/49.9 ms	[19]
CuZnS/GaN	350	0.36	8×10^{13}	0.14/40 ms	[45]
NiO/Ga ₂ O ₃	254	5.7×10^{-5}	5.45×10^9	0.34/3.65 s	[46]
$\text{SnO}_2/\text{1D-CsAg}_2\text{I}_3$	330	0.032	1.2×10^{11}	47/74 μs	This work

on CsAg_2I_3 [22], making it a promising self-powered UV PD. Future studies can focus on strategies to improve the mobility of CsAg_2I_3 and R , such as introducing a thin interfacial layer to reduce lattice mismatch and interfacial defects and improving the crystal orientation of CsAg_2I_3 . Furthermore, the performance dependence of the device on the UV intensity was also investigated. The increase in UV intensity shows a negative influence on R and D^* (Fig. 3d), consistent with other reports [47]. This influence is associated with the self-heating effect in the device at high illumination intensity, which may result in increases in carrier scattering and recombination probability.

To gain more insight into the UV-detection performance, the power absorption per unit volume (P_{abs}) and the electric field distributions of the $\text{SnO}_2/\text{CsAg}_2\text{I}_3$ heterojunction were theoretically investigated. Fig. 3e reveals that the P_{abs} distribution considerably depended on the wavelength, and < 300 nm incident light was absorbed by both layers, while 300–350 nm light was almost absorbed by CsAg_2I_3 . For long-wavelength light,

negligible P_{abs} were observed. The distribution is closely consistent with the bandgap alignment, which strongly suppresses the relatively high PPC effect of SnO_2 in the UV PDs under the working spectrum. Particularly, the electric field distribution for 330 nm is presented in Fig. 3f. The periodic distribution of the electric field is due to the interference inside the structure. In contrast to the electric field intensity inside SnO_2 , the intensity degrades sharply in CsAg_2I_3 owing to its strong light absorption.

In addition to the toxicity of lead, the instability of conventional lead halide perovskites is the main obstacle limiting their further applications in optoelectronic devices [48]. Typically, hybrid lead halide perovskites in moist air would decompose because of the effect of hygroscopic amine salts. Even if the resulting materials and devices are dried again, the chemical reaction process is irreversible [49]. To characterize the stability, UV PDs were operated under 330 nm radiation, and they exhibited excellent stability (Fig. 4a, b). The photocurrent under continuous light soaking was also tested. It exhibited negligible

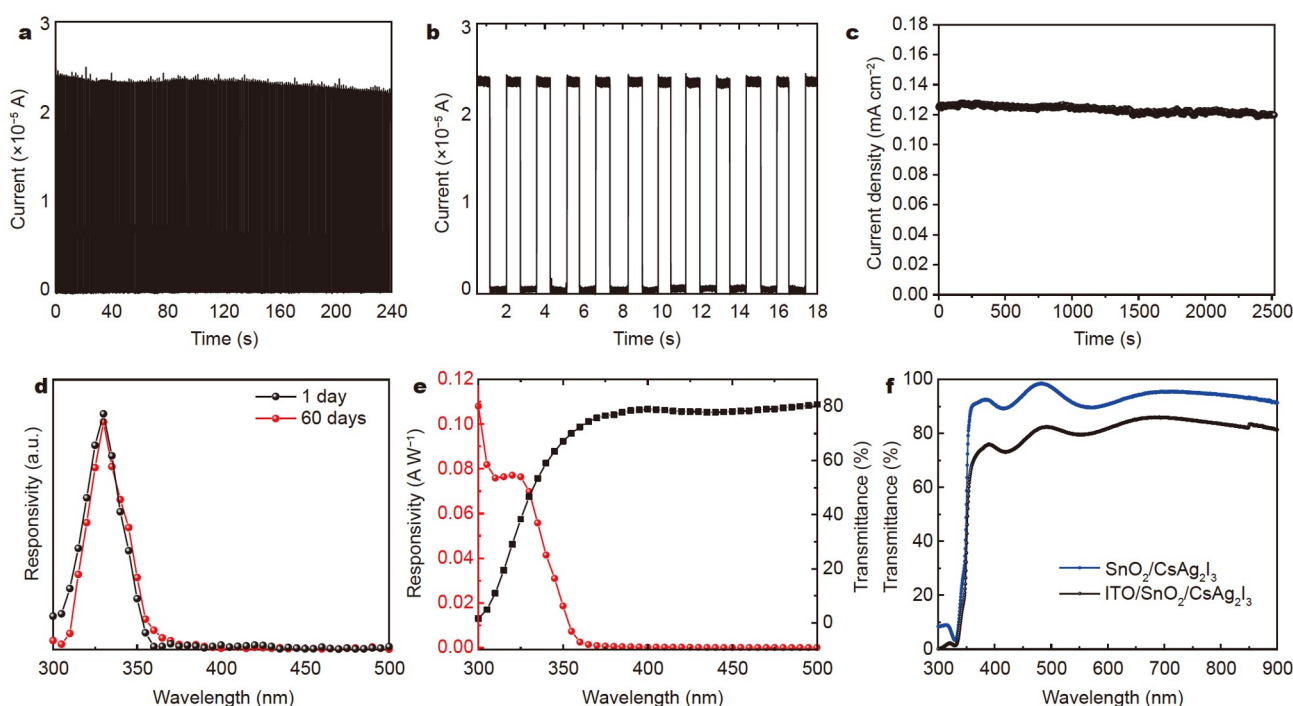


Figure 4 Responsivity evolution of SnO₂/1D-CsAg₂I₃ UV PDs in ambient air. (a, b) Time-dependent on/off cycling under 330 nm radiation (zero bias). (c, d) Photocurrent under continuous light soaking and photoresponsivity after 60 d. (e) Responsivity of UV PDs considering the available incident UV light of the heterojunction and the transmission of the device substrate. (f) Transmission of SnO₂/CsAg₂I₃ and ITO/SnO₂/CsAg₂I₃.

degradation after 2500 s (Fig. 4c). Moreover, the fabricated devices showed long-term stability after two-month storage (Fig. 4d). In this work, conventional ITO-glass was adopted as the substrate, which strongly absorbed UV radiation, particularly that with a short wavelength < 300 nm. To estimate the responsivity of the device by counting the available incident UV of the heterojunction, the transmission of the substrate was measured (Fig. 4e). A remarkable value of $\sim 0.1 \text{ A W}^{-1}$ was obtained. The UV loss caused by the substrate can be further optimized by choosing a substrate with high UV transmission, such as quartz-ITO [50]. In contrast to the case under UV light, the whole device (without top electrode) was highly transparent (up to 80% transparency) under light with a wavelength of > 360 nm. The value for the heterojunction (Fig. 4f) was > 90%, which indicates negligible influence on the nonworking spectrum, particularly under visible light. Fig. S6 shows a photograph of the as-prepared devices with and without the top silver electrode, which are highly transparent. All of these outstanding characteristics make the heterojunction a great choice for high-sensitive lead-free stable UV PDs, which are self-powered and visible-blind.

CONCLUSIONS

In summary, a high-quality CsAg₂I₃ film was successfully fabricated *via* a solution-processing method using pyridine as an additive to improve crystallization. The electronic structure revealed an intrinsic p-type behavior. The observed low exciton binding energy of 202 meV was comparable to the low-dimensional lead halide perovskite counterpart and thus favored the separation of photogenerated carriers. UV PDs based on all-solution-processed SnO₂/1D-CsAg₂I₃ n-p heterojunction were investigated, and 4.13 eV wide-bandgap SnO₂ with an extremely low Urbach energy of 0.214 eV was obtained to suppress the

PPC effect. The device shows an ultrafast photoresponse of 47/74 μs under zero bias. Additionally, the responsivity and detectability were as high as 0.032 A W^{-1} and 1.2×10^{11} Jones, respectively. Moreover, the device demonstrated remarkable stability under ambient conditions without encapsulation. Such high device performance can be attributed to the built-in electric field of the typical type-II heterojunction, the low exciton binding energy, and the suppressed PPC, which benefit carrier separation and transfer. Moreover, the remarkable high transparency (> 90%) of the heterojunction under light illumination with a wavelength of > 360 nm demonstrated its superior visible-blind characteristics. The stable nontoxicity of the 1D-CsAg₂I₃ material, the excellent device performance, and the visible-blind self-powered features reveal the potential of the proposed strategy for future UV PDs towards commercial applications.

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Conflict of interest The authors declare that they have no conflict of interest.

Supplementary information Supporting data are available in the online version of the paper.



Zhaosheng Hu received his PhD in engineering from the Graduate School of Life Science and Systems Engineering, Kyushu Institute of Technology, in 2018. He is currently an assistant professor at the School of Microelectronics, Xidian University. His research interests include lead-free metal halides, perovskite materials, photovoltaics, and related devices.



Yue Hao earned his doctoral degree in computational mathematics from Xi'an Jiaotong University in 1991. He is the vice president of Xidian University. His research interests include wide forbidden band semiconductor materials and devices, the reliability of ultradeep submicron small-sized devices, and SoC design and its design methodology.



Jingjing Chang received his PhD from the National University of Singapore in 2014. He was a research fellow in materials science & engineering at the National University of Singapore, from 2014 to 2015. He joined Xidian University, and now he is a professor. His research interests include perovskite materials, photovoltaics and organic electronics, and related devices.

基于全溶液法制备的 $\text{SnO}_2/\text{1D-CsAg}_2\text{I}_3$ 异质结的高灵敏自驱动可见光盲紫外探测器

胡赵胜^{1,2}, 张博尧¹, 张飞娟¹, 郭兴¹, 林珍华^{1,2}, 张进成¹, 郝跃¹, 常晶晶^{1,2*}

摘要 低维非铅金属卤化物材料因其优异的光电特性在可见光盲紫外探测领域具有很大的潜力. 本文通过吡啶添加剂辅助一步溶液法成功制备了高质量类钙钛矿 CsAg_2I_3 薄膜, 该薄膜展示出p型及202 meV低激子结合能(E_b). 其 E_b 甚至和2D铅卤钙钛矿相当, 有利于光生载流子分离. 此外, 探索了基于type-II型n-p异质结 $\text{SnO}_2/\text{CsAg}_2\text{I}_3$ 的自驱动紫外探测器, 其中4.13 eV宽带隙、0.214 eV低Urbach能量的 SnO_2 的引入抑制了持续光电导效应. 所制备的探测器件具有高灵敏特性及在可见光盲紫外探测器中超快的响应时间(47/74 μs), 大致快于当前 CsAg_2I_3 探测器数值的两个数量级. 其响应度和探测率分别高达0.032 A W^{-1} 和 1.2×10^{11} Jones. 此外, 异质结>90%高可见光透光性展现出优越的可见光盲特性. 这种优异的器件性能、可溶液法制备及长期稳定性为未来可见光盲、自驱动、稳定、环保的商用紫外探测器的制备提供了新思路.